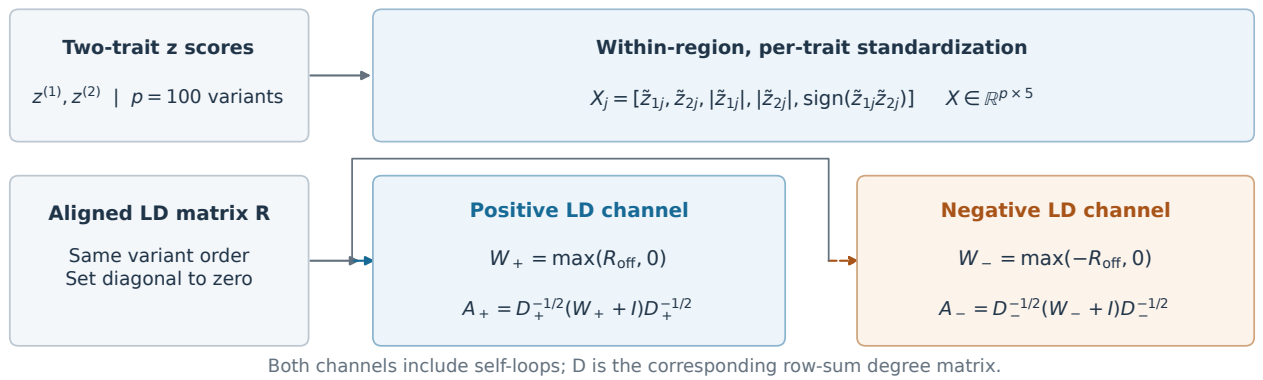
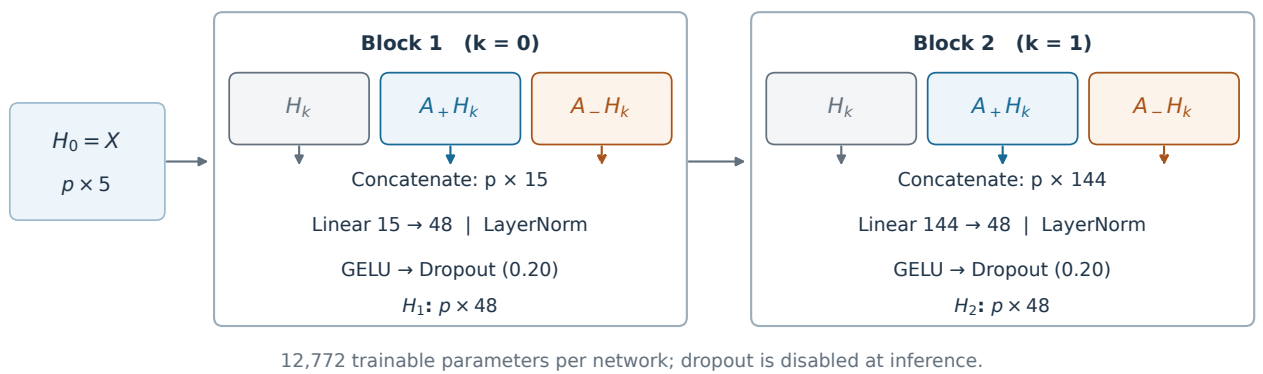


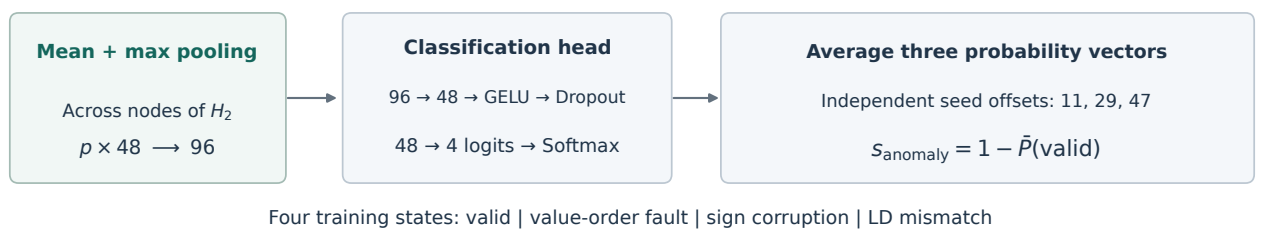
**a Aligned regional inputs and two signed LD operators**



**b Two message-passing blocks (same operators, different learned weights)**



**c Graph readout, classification and three-model ensemble**



**Warning only: no automatic repair, no eligibility override, no biological inference.**